Westermark and co-workers developed a unique method of revealing environmental history of natural rivers during the last 100–200 years. It is based on the freshwater pearl mussel *Margaritana margaritifera* living in these river. The mussel adds a new outer layer to its shell each year, incorporating into it trace metals (and also to some extent organics) dissolved in the water. The shell is cut and sectioned into small pieces, which are analyzed by α-track autoradiography, INAA and μ-PIXE. The latter is a "microprobe"-PIXE, i.e. the proton beam is focused by lenses to a resolution of a few microns. To get sufficient information the shell sample usually covers 3–15 years. With these techniques concentrations of some 30 elements have been determined retrospectively for >100 years in a number of Swedish rivers. Several unexplained trends have been observed, e.g. decreases in Ag, Au, Fe and Co. This is believed to be related to increasing air concentrations of SO_2. The observation of an increase in Br beginning in the 1940's is believed to be due to Br-additions to gasoline fuel. The technique has great potentials for increasing our knowledge on environmental changes of trace elements in nature. It could also be used as a long term biochemical control system at environmentally strategic places.

18.8. EXERCISES

18.1. The blood volume of a patient is to be determined by means of ^{32}P. For this purpose 15.0 ml of blood is withdrawn from the patient and mixed with a very small volume of $Na_2H^{32}PO_4$ of high specific activity. In 1 h the erythrocytes (red blood cells) take up all the ^{32}P; 1 ml is found to have an activity of 216 000 cpm in the detector system used. Exactly 5.00 ml of this tagged sample is reinjected into the patient, and 30 min later a new sample is withdrawn; 10 ml of this gives 2300 cpm. Calculate the blood volume.

18.2. A mixture of amino acids is to be assayed for cysteine. A 1.0 ml sample (density 1 g ml^{-1}) is withdrawn, and 2.61 mg of ^{35}S labelled cysteine of specific activity 0.862 μCi mg^{-1} is added. From this mixture pure cysteine is isolated by liquid-partition chromatography; 30.6 mg is isolated and measured to give 169 000 cpm in 27% detection efficiency. What is the percentage of cysteine in the original mixture?

18.3. In order to determine the lead content of a color pigment, 8.9871 g was dissolved in conc. HNO_3, and 5.00 ml ^{210}Pb solution added. After excess acid had been removed through evaporation, excess 1 M NaCl was added, the solution heated and filtered. After cooling and crystallization, the $PbCl_2$ was washed and recrystallized, 0.3276 g of the crystals was measured in a scintillation counter, giving 185 160 counts in 5 min. 1.000 ml of the original ^{210}Pb solution gave 57 000 cpm. The background was 362 cpm. Calculate the lead content (%) of the pigment.

18.4. What is the smallest amount of indium which can be determined in a 100 mg aluminum sample using NAA with a neutron flux of 10^{16} n m^{-2} s^{-1}? Consider the neutron capture in both ^{27}Al and ^{115}In: $^{27}Al(n,\gamma\ \sigma\ 0.230\ b)^{28}Al$ $t_{½}$ 2.25 min; $^{115}In(n,\gamma\ \sigma\ 45\ b)^{116}In$ $t_{½}$ 14 s. The lowest detectable activity for ^{116}In is assumed to be 10 Bq, and the interference from ^{28}Al not more than 20%.

18.5. In order to determine the amount of gallium in meteorite iron, 373.5 mg meteorite iron (A) and 10.32 mg gallium oxinate (B) were irradiated in a reactor under similar conditions in 30 min. After a short cooling, A was dissolved in concentrated HCl and 4.53 mg inactive Ga^{3+} was added. After a number of chemical separation steps, which were not quantitative, a precipitate of 25.13 mg pure gallium oxinate was isolated (C). Sample B was also dissolved and diluted to 50 ml; 0.50 ml was removed, 4 mg inactive Ga^{3+} added, and gallium oxinate precipitated (D). The radioactive decay curves gave two straight lines: $\log R_C = 3.401-0.0213\ t$, and $\log R_D = 3.445-0.0213\ t$. What was the gallium content in sample A?

18.6. A 10.0 g sample of iodobenzene is shaken with 100 ml of 1 M KI solution containing 2500 cpm ^{131}I. The activity of the iodobenzene at the end of 2 h is 250 cpm. What percent of the iodine atoms in the iodobenzene have exchanged with the iodide solution?

18.7. A sodium iodide solution contains some radioactive ^{129}I. An ethanol solution was prepared containing 0.135 M of this sodium iodide and 0.910 M inactive C_2H_5I. In the exchange reaction

$$C_2H_5I + {}^{129}I^- \underset{k_r}{\overset{k_f}{\rightleftarrows}} C_2H_5{}^{129}I + I^- \tag{18.39}$$

the reaction rate constant is assumed to be the same in both directions: $k_f = k_r$. One part (A) of the solution was removed and heated to high temperature so that equilibrium was rapidly reached. Another part (B) was kept in a thermostated bath at 30°C. After 50 min ethyl iodide was separated from both solutions. The concentration of radioactive iodine in C_2H_5I in B was found to be only 64.7% of that in A. Calculate k ($k_r = k\ a\ b$ in §18.4.2).

18.8. With Figure 18.9 one can estimate the stability constants ($\beta_1 - \beta_3$) for the lutetium acetylacetonate complexes. Make this estimation using Figure 18.9. A simplified approach to estimate β_n is the use of the approximate relations $k_n = \beta_n/\beta_{n-1} = 1/[\mathrm{Aa}]_{\overline{n}=n-0.5}$ and $\overline{n} = z-\mathrm{d}(\log D)/\mathrm{d}(\log[\mathrm{Aa}])$; for Lu^{3+} $n \leq 4$, but $z = 3$.

18.9. Calculate β_4 and the distribution constant using (18.32) and Figure 18.9.

18.10. One wants to determine the residual liquid volume of a closed sedimentation tank (nominal volume 80 m^3), which has been in use for many years, and in which $CaSO_4$ precipitates. 0.50 ml $^{24}Na_2SO_4$ (specific activity 3.2×10^8 cpm

ml^{-1}) is added to the tank, and 10 ml withdrawn after 2 h of settling; measurements yield a net value (background subtracted) of 500 counts in 10 min. Calculate the free liquid volume in the tank.

18.11. Calculate the critical deposition potential ($E - E^0$) for 10^{-12} M ^{210}Bi on a gold cathode (no over-voltage) from the Nernst equation (18.6), where the chemical activity of the reduced state (Bi^0) is set to unity.

18.12. A mineral ore contains cobalt and small amounts of nickel. In order to determine the nickel concentration it must be separated from cobalt. Solvent extraction using 0.01 M 8-hydroxyquinoline in $CHCl_3$ is chosen. Which metal should be extracted from the other, and at what pH? Consider Figure 18.3 and connected text.

18.13. In a solvent extraction system consisting of uranium and lanthanum in 1 M HNO_3 and 100% TBP, $D_U = 20$ and $D_{La} = 0.07$. If a phase ratio $\theta = V_{org} / V_{aq} = 0.5$ is chosen, how much uranium is removed from the aqueous phase in three repeated extractions? How much of the lanthanum is co-extracted? The fraction extracted with n fresh organic volumes (V_{org}) from one aqueous volume (V_{aq}) is:

$$E_n = 1 - (1 + D\theta)^{-n} \tag{18.40}$$

18.9. LITERATURE

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